

KABILAN N S

Digital Verification Engineer (Fresher) | Verilog | SystemVerilog | RTL Design & Verification

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PROFESSIONAL SUMMARY

Electronics and Communication Engineering graduate pursuing dedicated Digital Verification (DV) training at VLSI Guru, with hands-on project experience in Verilog and SystemVerilog RTL design and verification. Built and simulated multiple RTL designs including a low-power UART for SoC communication, a Gray-code SR flip-flop sequencer, a D flip-flop counter, and a memory module with self-checking testbenches. Comfortable with ModelSim/Questasim simulation flows and Linux-based development environments. Seeking an entry-level Digital Verification / RTL Design role to apply strong fundamentals in digital logic, functional verification, and embedded systems.

TECHNICAL SKILLS

- Languages: Verilog, SystemVerilog
- Verification & Design Concepts: RTL Design, Functional Verification, Testbench Development, Self-Checking Testbenches, Sequential & Combinational Logic, FSM Design
- Tools: ModelSim, Questasim, GVim, Arduino IDE
- Operating System: Linux
- Currently Building: Exposure to UVM and SystemVerilog Assertions (SVA) through VLSI Guru DV training

SOFT SKILLS

- Problem Solving • Teamwork & Collaboration • Time Management • Adaptability • Communication

EDUCATION

B.E., Electronics and Communication Engineering

2022 – 2026

University College of Engineering, Nagercoil, Tamil Nadu | Aggregate: 70%

Higher Secondary Certificate (HSC)

2022

V S Boys Government Aided Higher Secondary School, Tiruvarur, Tamil Nadu | Percentage: 65%

PROFESSIONAL EXPERIENCE

Embedded Systems Intern — KELTRON-IoT

Jul 2025 (4 weeks)

Thiruvananthapuram, Kerala

- Developed and tested an IoT-based embedded system during a 4-week internship, working on sensor integration and embedded firmware for real-time data handling.
- Debugged and validated hardware-software interaction on embedded hardware, strengthening practical skills in embedded C, sensor interfacing, and system testing.

ACADEMIC PROJECTS

Low Power UART for SoC Communication using ESP32

Final Year Project

- Designed a low-power UART communication system on ESP32 for efficient serial data transmission and reception between SoC peripherals, integrating low-power sleep modes to reduce idle-state power consumption.
- Validated reliable end-to-end data transfer using Arduino IDE and Serial Monitor, confirming protocol correctness under test conditions.

SR Flip-Flop – Gray Code Sequencer (Verilog)

- Designed and implemented an SR flip-flop-based sequencer in Verilog using Gray-code state transitions, ensuring only one bit changes between consecutive states for an efficient, glitch-resistant scheme.

Memory Design (Verilog)

- Designed a memory module in Verilog covering addressing, read/write operations, and data storage; verified functionality with a dedicated testbench, simulation waveforms, and targeted memory test cases.

D Flip-Flop Counter (Verilog)

- Built a counter using D flip-flops in Verilog to reinforce fundamentals of sequential circuit design and clocked logic.

LANGUAGES

- Tamil (Native), English (Professional Working Proficiency)